

Unit 2 - Linear Algebra Vectors

MSDS 520

Meharry Medical College

Outline

- 1 Why Vectors?
- 2 Vectors as Arrows, Rows, and Columns
- 3 Vector Operations
- 4 The Dot Product and the Angle
- 5 Linear Combinations, Span, and Independence
- 6 Basis and Change of Basis
- 7 Looking Ahead: Where Matrices Come From
- 8 Review

Why Vectors?

Where This Shows Up in Data Science

A row of a data table is a vector

- **Every record is a point.** One patient, one house, one image, one document becomes a list of numbers, and the whole dataset becomes a cloud of points.
- **Similarity is geometry.** Recommenders, search, and duplicate detection all rank things by **cosine similarity**, which is an angle.
- **Distance is the basis of clustering.** k -means, nearest neighbors, and anomaly detection all measure how far apart two vectors are.
- **Redundant features are dependent vectors.** When two columns carry the same information, their vectors lie on one line. That is **collinearity**, and it wrecks a regression.
- **Compression is choosing a better basis.** PCA, at the end of this unit, is exactly that.

Unit 2-1 wrote systems as grids of numbers. Here we ask what a **column** of numbers means on its own.

What We Will Build

The plan

- ① A **vector**, drawn as an arrow and written as a column (or row) of numbers.
- ② A vector's **magnitude**; **adding**, **scaling**, and **subtracting** vectors; the **distance** between points; and **unit vectors**.
- ③ The **dot product**: one number that measures how much two vectors agree. From it: the **angle**, **cosine similarity** (how a spam filter tells which emails are alike), **orthogonality**, and **projection**.
- ④ **Linear combinations**, **span**, and **independence**: which points you can reach, and whether anything is redundant.
- ⑤ **Basis**, **dimension**, **change of basis**, and a look ahead to **matrices**.

Notebooks: U2-2_Vectors-1_ArrowsAndOps, -2_DotProductAndAngle, -3_ProjectionAndSimilarity, -4_LinearCombinationsAndSpan, -5_IndependenceAndBasis.

Vectors as Arrows, Rows, and Columns

Two Pictures of the Same Thing

Definition

A **vector** is a quantity with both a **magnitude** (how big) and a **direction** (which way).

Size only

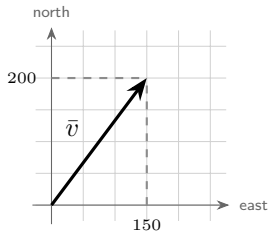
Size and direction: a vector

distance: 5 km **displacement:** 5 km north

speed: 250 m/s **velocity:** 150 east + 200 north (m/s)

A plane's velocity is **drawn** as an arrow and **written** as components, one per axis:

$$\vec{v} = \begin{bmatrix} 150 \\ 200 \end{bmatrix} \begin{array}{l} \leftarrow \text{east} \\ \leftarrow \text{north} \end{array}$$



Velocity in m/s; one square is 50 m/s.

Notation: marking a vector

- $\mathbf{v}$, $\vec{v}$, $\hat{v}$ all mean “vector”: bold when typing, bar or arrow by hand (we use the **bar**).
- $\hat{v}$ (hat) means a vector of length exactly 1, a **unit vector**.

Notebook U2-2_Vectors-1_ArrowsAndOps, §1, the arrow and the column side by side

Three Running Examples

Three vectors will follow us through the whole lecture:

Velocity

A plane flying 150 m/s east and 200 m/s north:

$$\vec{v} = \begin{bmatrix} 150 \\ 200 \end{bmatrix}$$

One number per direction.

Patient

Age, systolic blood pressure, A1c:

$$\vec{p} = \begin{bmatrix} 54 \\ 130 \\ 6.1 \end{bmatrix}$$

One number per measurement.

Email

Counts of *free*, *win*, *cash*, *meeting*:

$$\vec{d}_1 = \begin{bmatrix} 2 \\ 1 \\ 2 \\ 0 \end{bmatrix}$$

One number per word.

The question they build toward

- Two patients, or two emails: **how similar are they?** Is a new email **spam**?
- By the end, “similar” is a number we can **compute**: how **close** two vectors are, or how nearly they **point the same way**.

Turning an Email into a Vector

A **spam filter** compares emails, so first it turns each one into numbers.

Algorithm: a count vector

- ① Fix a **vocabulary**: the words to track.
- ② In each email, **count** each word.
- ③ The list of counts **is** the vector.

- Order and capitals are thrown away: *Free* counts as *free*.
- Real filters track thousands of words: $\mathbb{R}^{5000}$ or more.

	$\bar{d}_1$	$\bar{d}_2$	$\bar{d}_3$
<i>free</i>	2	4	2
<i>win</i>	1	2	0
<i>cash</i>	2	4	1
<i>meeting</i>	0	0	2

Each **column** is one email's vector.

Three emails

$\bar{d}_1$	"Win cash now! Free cash, free entry."	spam
$\bar{d}_2$	"Win cash now! Free cash, free entry. Win free cash, free prizes, cash today."	spam
$\bar{d}_3$	"Free lunch at the meeting. After the meeting, free parking; bring cash."	not spam

Columns and Rows: Two Ways to Write a Vector

Column or row: same numbers, different shape

$$\bar{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \text{ (column, } 2 \times 1) \qquad \bar{v} = [3 \ 4] \text{ (row, } 1 \times 2)$$

- Same information, **different shapes**: $n \times 1$ or $1 \times n$. From Unit 2-3 on, the shape decides what can be multiplied by what.
- We write vectors as **columns by default**; a patient stored in a data table is a **row**.
- A **point** is neither: it stays a tuple, $(3, 4)$.

In numpy: three kinds of array

Shape	Built with	What it is
$(n,)$	<code>np.array([3, 4])</code>	a flat list, no orientation
$(n,1)$	<code>np.array([[3], [4]])</code>	a column : n rows, 1 entry each
$(1,n)$	<code>np.array([[3, 4]])</code>	a row : 1 row, n entries

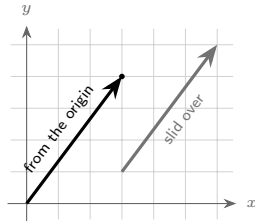
Check with `v.shape`. Most code uses the flat $(n,)$ kind.

Point Versus Vector

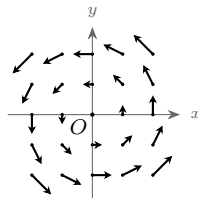
Same numbers, different ideas

- A **point** $(3, 4)$ is a **location**. It sits where it sits.
- A **vector** $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$ is a **displacement**: 3 across, 4 up. It has **no home**.

- We draw vectors from the origin **by default**, so arrow and point look alike.
- Sliding an arrow, same length and direction, **changes nothing**.
- Addition, next section, relies on that freedom.



Two drawings of the **same** vector. Only the dot is a point.



A **velocity field**: fluid swirling around the origin O . Each arrow sits **where the fluid is**, and the fluid moves faster farther out.

Beyond the Plane

Nothing about the definition needs two components. A vector with n components lives in $\mathbb{R}^n$, the set of all lists of n real numbers.

$$\bar{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{R}^n$$

The running examples, in any dimension

- **Velocity**: add “up” for a plane that is climbing, and it lives in $\mathbb{R}^3$.
- **Patient**: 3 measurements is $\mathbb{R}^3$; a 40-value lab panel is $\mathbb{R}^{40}$.
- **Email**: a 5000-word vocabulary puts every email in $\mathbb{R}^{5000}$.

The habit to build

- You cannot **picture** $\mathbb{R}^{40}$, and you do not need to.
- Every formula here is a sum over components, so it works for **any** n .
- Reason with the picture in $\mathbb{R}^2$; trust the algebra in $\mathbb{R}^n$.

Vector Operations

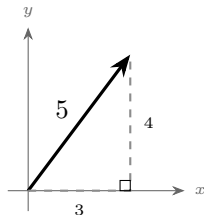
Magnitude: How Long Is a Vector?

Definition

The **magnitude**, or **norm**, of $\vec{v} \in \mathbb{R}^n$:

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

Never negative; zero only for $\vec{0}$.



Legs 3 and 4, hypotenuse 5.

Running examples

- **Velocity:** $\|\vec{v}\| = \sqrt{150^2 + 200^2} = 250$ m/s, the plane's **speed**.
- **Email:** $\|\vec{d}_1\| = \sqrt{4 + 1 + 4 + 0} = 3$ and $\|\vec{d}_2\| = 6$: longer email, longer vector.
- **Patient:** $\|\vec{p}\|$ mixes years, mmHg, and percent, so it is **meaningless**.

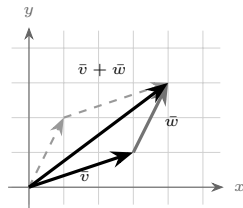
Notebook U2-2_Vectors-1_ArrowsAndOps, §5, magnitude as a hypotenuse

Addition: Head to Tail

Componentwise, as you would guess:

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

- Tail of $\vec{w}$ on the head of $\vec{v}$.
- The sum: first tail to last head.
- Either order: $\vec{v} + \vec{w} = \vec{w} + \vec{v}$.



Either route reaches (4,3).

Running examples

- **Velocity:** a plane's velocity through the air $\vec{v}$, plus the wind $\vec{w}$, is its velocity over the **ground**, $\vec{v} + \vec{w}$.
- **Email:** paste $\vec{d}_1$ and $\vec{d}_3$ together and the counts add: 4 *free*, 1 *win*, 3 *cash*, 2 *meeting*.

The triangle inequality

$$\|\vec{v} + \vec{w}\| \leq \|\vec{v}\| + \|\vec{w}\|$$

Here $5 \leq \sqrt{10} + \sqrt{5} \approx 5.4$: the direct route wins.

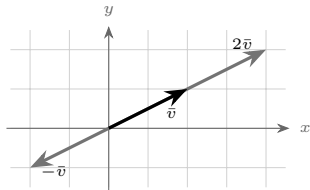
Notebook U2-2_Vectors-1_ArrowsAndOps, §2, addition drawn head to tail

Scalar Multiplication: Stretch, Shrink, Flip

A **scalar** is a plain number. Multiplying by it **scales**:

$$c \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} c v_1 \\ c v_2 \end{bmatrix}$$

- $c > 1$ stretches; $0 < c < 1$ shrinks.
- $c < 0$ flips; $c = 0$ gives $\vec{0}$.
- Every multiple stays on **one line through the origin**: remember this for **span**.



All multiples of $\vec{v}$ lie on one line.

Running examples

- **Velocity**: $2\vec{v}$ doubles the speed on the same heading; $-\vec{v}$ reverses course.
- **Units**: m/s to km/h is scaling by 3.6, so 250 m/s is 900 km/h.
- **Email**: $\vec{d}_2 = 2\vec{d}_1$: every word twice as often. Same **direction**, twice the length.

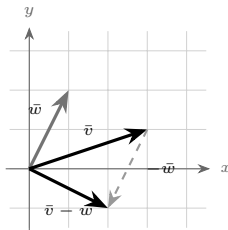
Subtraction: Add the Reverse

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

Addition in disguise

$$\bar{v} - \bar{w} = \bar{v} + (-\bar{w})$$

- $-\bar{w}$: same length, opposite way.
- Put its tail on the head of $\bar{v}$.



$\bar{v}$, then the reversed $\bar{w}$ (dashed).

Velocity: a plane in a headwind

A plane flies 250 m/s east; a 50 m/s **headwind** takes some away:

$$\begin{bmatrix} 250 \\ 0 \end{bmatrix} - \begin{bmatrix} 50 \\ 0 \end{bmatrix} = \begin{bmatrix} 200 \\ 0 \end{bmatrix}$$

Its velocity over the **ground**.

Patient: what changed

A year later, minus today:

$$\begin{bmatrix} 55 \\ 136 \\ 6.5 \end{bmatrix} - \begin{bmatrix} 54 \\ 130 \\ 6.1 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \\ 0.4 \end{bmatrix}$$

$\bar{p}' - \bar{p}$: up 1 year, 6 mmHg, 0.4 A1c.

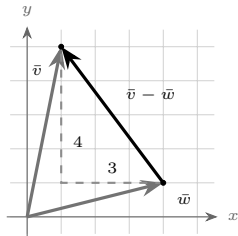
Distance Between Two Points

Definition

The **distance** between two points is the length of the trip between them:

$$d(\bar{v}, \bar{w}) = \|\bar{v} - \bar{w}\| = \sqrt{\sum_i (v_i - w_i)^2}$$

Euclidean distance: what k -means and nearest-neighbor methods measure.



Legs 3 and 4: the points are 5 apart.

Running example: $\bar{p}$ and a second patient $\bar{q}$, and a warning

$$\bar{p} - \bar{q} = \begin{bmatrix} 54 \\ 130 \\ 6.1 \end{bmatrix} - \begin{bmatrix} 58 \\ 138 \\ 6.4 \end{bmatrix} = \begin{bmatrix} -4 \\ -8 \\ -0.3 \end{bmatrix}, \quad d(\bar{p}, \bar{q}) = \sqrt{16 + 64 + 0.09} \approx 8.95$$

- Blood pressure supplies 64 of the 80; A1c supplies 0.09 and barely registers.
- The biggest units win, so **standardize** before measuring distance.

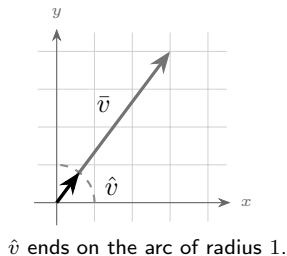
Unit Vectors: Getting $\hat{v}$ from $\bar{v}$

A **unit vector** has length exactly 1; the **hat** marks one. $\hat{v}$ keeps the **direction** of $\bar{v}$ and drops its size.

Normalizing: divide by the length

$$\hat{v} = \frac{1}{\|\bar{v}\|} \bar{v} = \frac{\bar{v}}{\|\bar{v}\|}$$

- **Scaling** by $1/\|\bar{v}\|$: same direction, length 1.
- Not defined for $\bar{0}$, which has no direction.



Running examples

- **Velocity**: the **heading** is $\hat{v} = \frac{1}{250} \begin{bmatrix} 150 \\ 200 \end{bmatrix} = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$; check $0.6^2 + 0.8^2 = 1$.
- **Email**: $\bar{d}_2 = 2\bar{d}_1$, so $\hat{d}_2 = \hat{d}_1$: same mix of words, length forgotten.

The Standard Unit Vectors

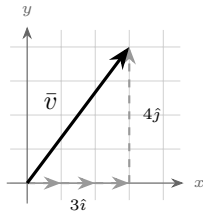
One step along each axis:

$$\hat{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \hat{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Scale and add them to build **every** vector:

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix} = 3\hat{i} + 4\hat{j}$$

The components are the **coefficients** of $\hat{i}$ and $\hat{j}$: a **choice of axes**, which we will later change.



Running examples

- **Velocity:** $150\hat{i} + 200\hat{j}$ reads “150 m/s east plus 200 m/s north.”
- **Email:** each word gets its own unit vector, an email saying just that word once. The counts are the coefficients.

Notebook U2-2_Vectors-1_ArrowsAndOps, §4, building $\bar{v}$ from $\hat{i}$ and $\hat{j}$

Exercise: Vector Arithmetic

Try it

Let $\bar{u} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\bar{v} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$.

- ① Compute $\bar{u} + \bar{v}$ and $\bar{u} - \bar{v}$.
- ② Compute $2\bar{u} - 3\bar{v}$.
- ③ Compute $\|\bar{u}\|$.
- ④ Find the distance between the points at the heads of $\bar{u}$ and $\bar{v}$.
- ⑤ Write $\bar{u}$ in terms of $\hat{i}$ and $\hat{j}$.
- ⑥ What goes wrong if you try to add $\bar{u}$ to $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$?

Exercise: Solution

Answers

- ① $\bar{u} + \bar{v} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$, $\bar{u} - \bar{v} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. Add and subtract component by component.
- ② $2\bar{u} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$ and $3\bar{v} = \begin{bmatrix} 12 \\ -3 \end{bmatrix}$, so $2\bar{u} - 3\bar{v} = \begin{bmatrix} -10 \\ 9 \end{bmatrix}$. Scale first, then subtract.
- ③ $\|\bar{u}\| = \sqrt{1^2 + 3^2} = \sqrt{10} \approx 3.16$. Leave it exact unless a decimal is asked for.
- ④ From part 1, $\bar{u} - \bar{v} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$, so the distance is $\sqrt{(-3)^2 + 4^2} = \sqrt{25} = 5$.
A 3-4-5 triangle again.
- ⑤ $\bar{u} = 1\hat{i} + 3\hat{j}$.
- ⑥ **Nothing is defined.** $\bar{u}$ lives in $\mathbb{R}^2$, the other in $\mathbb{R}^3$: no third component of $\bar{u}$ to add to the 3. Operations need **matching dimensions**; in `numpy` this raises a broadcast error.

The Dot Product and the Angle

One Number That Compares Two Vectors

Definition

For $\bar{v}, \bar{w} \in \mathbb{R}^n$, the **dot product** (or **inner product**) is

$$\bar{v} \cdot \bar{w} = v_1 w_1 + v_2 w_2 + \cdots + v_n w_n = \sum_{i=1}^n v_i w_i$$

How to read it

- Multiply matching components, then add.
- The result is a **number**, not a vector.
- Both need the **same** number of components.

Running example: two emails

$$\bar{d}_1 \cdot \bar{d}_3 = (2)(2) + (1)(0) + (2)(1) + (0)(2) = 6$$

- Each word adds (count here) $\times$ (count there).
- Emails with **no words in common** score 0.

Notebook U2-2_Vectors-2_DotProductAndAngle, §1, the dot product by hand and with `np.dot`

Careful: Which Multiplication?

“Multiply two vectors” is ambiguous: the **symbol** decides

Written	Name	Gives
$c \bar{v}$	scalar multiplication	a vector (each entry times c)
$\bar{v} \cdot \bar{w}$	dot product	a number
$\bar{v} \times \bar{w}$	cross product	a vector , only in $\mathbb{R}^3$ (not in this course)
$\bar{v} \bar{w}$	side by side	undefined for two column vectors

Be careful

- Side by side, $\bar{v} \bar{w}$, is **not** the dot product. Write the dot.
- Side by side gets a meaning in Unit 2-3, for matrices.

In code, with the emails

- `d1 @ d3` is the dot product: 6.
- `d1 * d3` is **not**: it multiplies entry by entry, `[4 0 2 0]`.

Rules, and What the Sign Tells You

Properties

- $\bar{v} \cdot \bar{w} = \bar{w} \cdot \bar{v}$
- $\bar{u} \cdot (\bar{v} + \bar{w}) = \bar{u} \cdot \bar{v} + \bar{u} \cdot \bar{w}$
- $(c\bar{v}) \cdot \bar{w} = c(\bar{v} \cdot \bar{w})$
- $\bar{v} \cdot \bar{v} = \|\bar{v}\|^2 \geq 0$

The last rule is the link to **length**:

$$\|\bar{v}\| = \sqrt{\bar{v} \cdot \bar{v}}$$

The sign is a verdict on direction

- **Positive:** they broadly **agree**, within a quarter turn.
- **Zero: perpendicular.** Neither has any component along the other.
- **Negative:** they broadly **oppose**, more than a quarter turn apart.

Running example: a plane flying along $\bar{v}$, in a wind $\bar{w}$

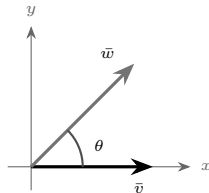
- $\bar{v} \cdot \bar{w} > 0$: part **tailwind**. $\bar{v} \cdot \bar{w} < 0$: part **headwind**.
- $\bar{v} \cdot \bar{w} = 0$: pure **crosswind**. No gain or loss along the heading, but it still pushes the plane **sideways**.

The Angle Between Two Vectors

The key formula

$$\cos \theta = \frac{\vec{v} \cdot \vec{w}}{\|\vec{v}\| \|\vec{w}\|}$$

- Dividing by both lengths leaves **direction alone**.
- The result is always between -1 and 1 .



Worked: a plane heading east, wind blowing northeast

$$\vec{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \vec{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} : \quad \cos \theta = \frac{1}{1 \cdot \sqrt{2}} = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$$

Running examples, beyond the picture

- **Email:** for $\vec{d}_1, \vec{d}_3$, $\cos \theta = 6/(3 \cdot 3) = 2/3$, so $\theta \approx 48^\circ$.
- **Patient:** no picture in $\mathbb{R}^{40}$, but the formula still returns a number in $[-1, 1]$.

Cosine Similarity

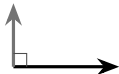
Definition

The **cosine similarity** of nonzero $\bar{v}, \bar{w}$ is the cosine of the angle between them:

$$\cos \theta = \frac{\bar{v} \cdot \bar{w}}{\|\bar{v}\| \|\bar{w}\|}$$



$\cos \theta = +1$
same direction



$\cos \theta = 0$
perpendicular, nothing shared



$\cos \theta = -1$
opposite directions

Why it gets its own name

- It sees **direction only**: scaling either vector leaves it unchanged.
- That is exactly right when size is noise, like the **length** of an email.

Notebook U2-2_Vectors-3_ProjectionAndSimilarity, §3, a patient similarity heatmap

Why Spam Filters Compare by Cosine

The three emails

$\bar{d}_1$	"Win cash now! Free cash, free entry."	spam
$\bar{d}_2$	"Win cash now! Free cash, free entry. Win free cash, free prizes, cash today."	spam
$\bar{d}_3$	"Free lunch at the meeting. After the meeting, free parking; bring cash."	not spam

	$\bar{d}_1$	$\bar{d}_2$	$\bar{d}_3$
<i>free</i>	2	4	2
<i>win</i>	1	2	0
<i>cash</i>	2	4	1
<i>meeting</i>	0	0	2

$$\|\bar{d}_1\| = \|\bar{d}_3\| = 3, \quad \|\bar{d}_2\| = 6$$

Which email is most like the spam $\bar{d}_1$?

- **Distance** picks $\bar{d}_3$, not spam: fooled by length.
- **Cosine** picks $\bar{d}_2$: the same spam, longer.

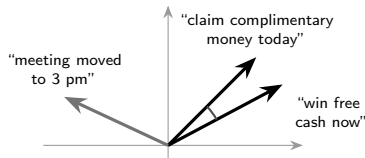
The numbers

	Distance from $\bar{d}_1$	Cosine with $\bar{d}_1$
$\bar{d}_2$ (spam, longer)	$\sqrt{2^2 + 1^2 + 2^2 + 0^2} = 3$	$(8 + 2 + 8 + 0)/(3 \cdot 6) = 1$
$\bar{d}_3$ (not spam)	$\sqrt{0^2 + 1^2 + 1^2 + 2^2} \approx 2.45$	$(4 + 0 + 2 + 0)/(3 \cdot 3) \approx 0.67$

Beyond Counting Words: Embeddings

The problem with counting words

- “Win free cash now” and “Claim complimentary money today” mean the **same thing**.
- No shared words, so the count vectors are perpendicular: $\cosine\ 0$.
- Counts see **spelling**, not **meaning**.



The goal: same meaning, **small angle**.

What we want instead

- “cat” and “dog” should come out **similar**; “cat” and “agglomerative” should not.
- Counts cannot do it: each word is its own unit vector, so **any** two different words are perpendicular.
- We need vectors built from **meaning**. Those are **embeddings**.

What an Embedding Looks Like

The idea: an embedding

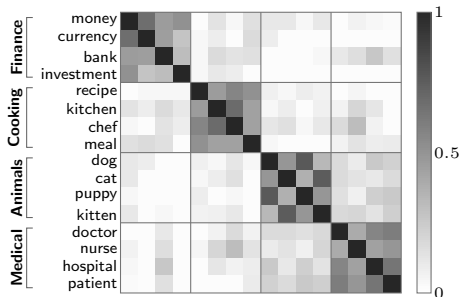
- Each word becomes a vector of **learned** numbers, not counts: hundreds to thousands of components.
- A model learns them by **predicting each word from the words around it**.
- Trained so that **similar meanings point the same way**.

A sentence, one column per word: “The cat sat on the mat”

	<i>the</i>	<i>cat</i>	<i>sat</i>	<i>on</i>	<i>the</i>	<i>mat</i>
entry 1	0.23	0.32	0.23	-2.63	0.23	0.01
2	0.51	-1.47	0.00	-0.77	0.51	1.99
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$n - 1$	-0.21	0.86	-1.47	0.23	-0.21	0.23
n	1.99	0.10	0.51	0.11	1.99	0.50

- Each column is one word's **vector**.
- Both copies of *the* get the **same** column: the vector belongs to the word.
- Side by side, the columns form a grid of numbers, a **matrix**: Unit 2-3.

Cosine Similarity Between Word Embeddings



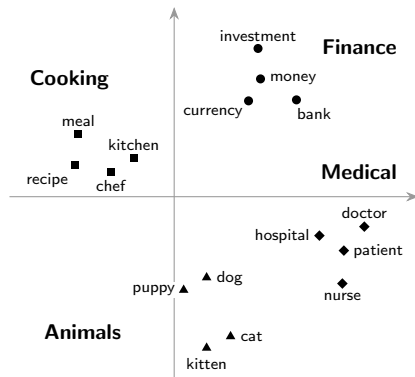
Columns are the same 16 words, in the same order.

How to read it

- Each cell is the **cosine similarity** of two words' embeddings; **darker** is more similar.
- The diagonal is 1: every word with itself.
- Dark 4×4 blocks**: words on the same topic point nearly the same way, a **small angle**.
- Light elsewhere**: words on different topics are **nearly perpendicular**.

The model was never told the topics. The blocks come only from how the words are used in text.

Similar Meanings Cluster Together



The same 16 embeddings, flattened to 2-D.

What the picture shows

- Each word's hundreds of numbers are **flattened to 2** so we can draw them, using **PCA** (the end of this unit) plus a method called t-SNE.
- Each topic lands in its own **cluster**: similar meaning, nearby points.
- The axes carry no units; only **who is near whom** matters.

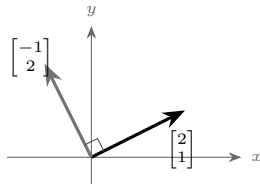
Back to the spam filter

Whole sentences get embeddings too. Now "win free cash now" and "claim complimentary money today" point nearly the **same way**, and the **same cosine** finds it. Search engines rank results this way.

Orthogonality

Definitions

- **Orthogonal** (perpendicular): $\vec{v} \cdot \vec{w} = 0$.
- **Orthonormal**: orthogonal **and** both unit vectors, $\|\vec{v}\| = \|\vec{w}\| = 1$.
- **Why**: $\cos 90^\circ = 0$.
- **Check**: $\begin{bmatrix} 2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 2 \end{bmatrix} = -2 + 2 = 0$.
- **Not** orthonormal: each has length $\sqrt{5}$.
Normalize both and they are, like $\hat{i}$, $\hat{j}$.
- No roots or trig needed: cheap even in $\mathbb{R}^{500}$.



Running examples

- **Velocity**: a pure **crosswind** adds nothing along the plane's heading; it only pushes the plane sideways.
- **Email**: two emails with **no words in common** are orthogonal.

Exercise: Dot Products and Angles

Try it

Let $\bar{a} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$, $\bar{b} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$, $\bar{c} = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$.

- ① Compute $\bar{a} \cdot \bar{b}$ and the angle between $\bar{a}$ and $\bar{b}$.
- ② Compute $\bar{a} \cdot \bar{c}$. What does the answer say about them?
- ③ Compute $\bar{b} \cdot \bar{c}$. What does the **sign** say?
- ④ Turn $\bar{b}$ into a unit vector.

Exercise: Solution

Answers

- ① $\bar{a} \cdot \bar{b} = (3)(3) + (0)(3) = 9$. With $\|\bar{a}\| = 3$ and $\|\bar{b}\| = \sqrt{18} = 3\sqrt{2}$,
 $\cos \theta = \frac{9}{3 \cdot 3\sqrt{2}} = \frac{1}{\sqrt{2}}$, so $\theta = 45^\circ$. ($\bar{a}$ lies along the x -axis and $\bar{b}$ on the diagonal, so this is the answer the picture predicts.)
- ② $\bar{a} \cdot \bar{c} = (3)(0) + (0)(-2) = 0$: they are **orthogonal**. One runs along the x -axis, the other along the y -axis.
- ③ $\bar{b} \cdot \bar{c} = (3)(0) + (3)(-2) = -6$. **Negative**, so the two point more than 90° apart: $\bar{b}$ heads up and right while $\bar{c}$ heads straight down.
- ④ $\hat{b} = \frac{1}{3\sqrt{2}} \begin{bmatrix} 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \approx \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}$, and $0.5 + 0.5 = 1$ as required.

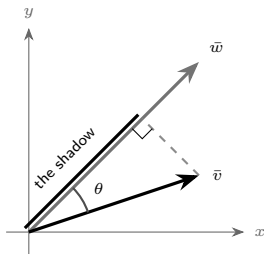
Projection: the Shadow of One Vector on Another

The **shadow** of $\vec{v}$ on the line through $\vec{w}$: the part of $\vec{v}$ running **along** $\vec{w}$.

Velocity: wind $\vec{v}$ on a plane heading northeast along $\vec{w}$, **not** along an axis:

$$\vec{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad \vec{w} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}, \quad \hat{w} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

($\hat{w}$ is the unit vector from the last exercise.)



1. Dot with $\hat{w}$: a **number**

$$\text{comp}_{\vec{w}}(\vec{v}) = \vec{v} \cdot \hat{w} = \|\vec{v}\| \cos \theta$$

The **component** of $\vec{v}$ along $\vec{w}$: the shadow's **length**. Here $\frac{3}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2} \approx 2.83$: the **tailwind**.

Only **part** of the wind: the rest, the dashed drop, is the **crosswind** (next slide).

Notebook U2-2_Vectors-3_ProjectionAndSimilarity, §1, the drop drawn with its right angle

2. Times $\hat{w}$: a **vector**

$$\text{proj}_{\vec{w}}(\vec{v}) = (\vec{v} \cdot \hat{w}) \hat{w}$$

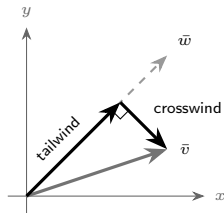
The **projection**: that length, pointed along $\vec{w}$. Here $2\sqrt{2} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$: the roots cancel.

The Leftover: the Crosswind

3. Subtract: the **leftover**

$$\bar{v} - \text{proj}_{\bar{w}}(\bar{v}) = \begin{bmatrix} 3 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

The part of the wind **across** the heading. It adds nothing to the speed along $\bar{w}$; it only shoves the plane **sideways**, off course.



True for **every** projection, not just this one

- **Perpendicular to $\bar{w}$:** $(1)(3) + (-1)(3) = 0$, a right angle.
- **The pieces rebuild $\bar{v}$:** $\begin{bmatrix} 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, tailwind plus crosswind.
- **Pythagoras:** $\text{tail}^2 + \text{cross}^2 = (2\sqrt{2})^2 + (\sqrt{2})^2 = 10 = \|\bar{v}\|^2$.

Look ahead: fitting a line to data (Unit 3) leaves errors that are this same kind of leftover, at a right angle to the fit.

Exercise: Projection and Similarity

Try it

- ① Project $\bar{v} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ onto $\bar{w} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$. Sketch it.
- ② Two patients are recorded as $\bar{p} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\bar{q} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$. What is their cosine similarity? Answer without computing anything.
- ③ Their Euclidean **distance**, on the other hand, is not small. Which of the two measures is the right one, and what does the choice depend on?

Exercise: Solution

Answers

- ① $\|\bar{w}\| = 3$, so $\hat{w} = \frac{1}{3} \begin{bmatrix} 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\bar{v} \cdot \hat{w} = (2)(1) + (4)(0) = 2$, so $\text{proj}_{\bar{w}}(\bar{v}) = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$. Since $\bar{w}$ lies along the x -axis, the projection just keeps the x -component of $\bar{v}$ and drops the y -component, which is what the sketch shows.
- ② **Exactly** 1. $\bar{q} = 3\bar{p}$, so the two vectors point along the same line in the same direction and the angle between them is 0. No arithmetic needed: scaling a vector never changes its direction, and cosine similarity sees direction only.
- ③ **It depends on whether size carries meaning.** If the components are counts (words in a document, visits in a year) then $\bar{q}$ is a bigger version of the same profile, and **cosine** is right. If they are clinical measurements, where a value three times higher is a genuinely different patient, then **distance** is right. Choosing between them is a modelling decision, not a mathematical one.

Linear Combinations, Span, and Independence

Linear Combinations

Definition

A **linear combination** of $\bar{v}_1, \dots, \bar{v}_k$ scales each one and adds:

$$c_1\bar{v}_1 + c_2\bar{v}_2 + \dots + c_k\bar{v}_k$$

The c_i are the **coefficients** or **weights**.

The two questions that matter

- ① **Which points** can you reach?
- ② For a given point, **which weights** get you there?

Running examples: we have been doing this all along

- **Velocity:** $\bar{v} = 150\hat{i} + 200\hat{j}$, weights 150 and 200.
- **Email:** the word counts are the weights on the word unit vectors.
- **Patient:** the **average** of $\bar{p}$ and $\bar{q}$ has weights $\frac{1}{2}, \frac{1}{2}$:

$$\frac{1}{2} \begin{bmatrix} 54 \\ 130 \\ 6.1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 58 \\ 138 \\ 6.4 \end{bmatrix} = \begin{bmatrix} 56 \\ 134 \\ 6.25 \end{bmatrix}$$

Span: Everything You Can Reach

Definition

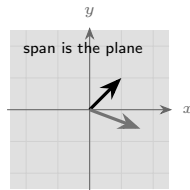
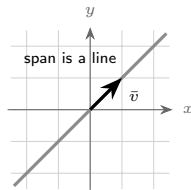
The **span** of a set of vectors is the set of **all** their linear combinations.

In the plane there are three possibilities:

- **One nonzero vector:** a **line** through the origin.
- **Two, in different directions:** the **whole plane**.
- **Two on the same line:** only **that line**.

Every span contains $\vec{0}$: set all weights to zero.

Notebook U2-2_Vectors-4_LinearCombinationsAndSpan, §2, the span filled in point by point



Finding the Weights Is Solving a System

Is $\bar{u} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ a combination of $\bar{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\bar{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$? Write out what that would mean:

$$a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \iff \begin{bmatrix} a+b \\ b \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \iff \begin{cases} a+b=3 \\ b=2 \end{cases}$$

Solve it

- One **row** at a time: a Unit 2-1 system.
- Row 2: $b = 2$. Row 1:
 $a = 3 - 2 = 1$.
- So $\bar{u} = 1\bar{v} + 2\bar{w}$.

Check: $\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$.

The three outcomes, as geometry

- **One solution:** the weights are **unique**.
- **No solution:** $\bar{u}$ lies **outside** the span.
- **Infinitely many:** the vectors are redundant, so many different linear combinations give the same $\bar{u}$.

Independence: Is Anything Redundant?

Definition: linearly dependent

A set of vectors is **linearly dependent** if one vector is a linear combination of the others. Two equivalent forms; **either works**:

$$\bar{u} = a\bar{v} + b\bar{w} \quad \text{or} \quad a\bar{u} + b\bar{v} + c\bar{w} = \bar{0} \text{ with } a, b, c \text{ not all } 0$$

- The first names the **redundant** vector, $\bar{u}$.
- The second treats all three alike (rearrange the first: weights $1, -a, -b$).
- “Not all 0” matters: all-zero weights give $\bar{0}$ for **any** vectors.

Definition: linearly independent

Not dependent: **no** vector is a combination of the others. Equivalently, $a\bar{u} + b\bar{v} + c\bar{w} = \bar{0}$ **only** when $a = b = c = 0$.

Worth memorizing: in $\mathbb{R}^n$, any **more than** n vectors are dependent.

Example: Testing for Dependence

Is $\bar{u}$ a combination of $\bar{v}$ and $\bar{w}$? Read the components one row at a time:

$$a \underbrace{\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}}_{\bar{v}} + b \underbrace{\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}}_{\bar{w}} = \underbrace{\begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}}_{\bar{u}} \quad \longleftrightarrow \quad \begin{cases} a = 2 \\ 2a + b = 3 \\ b = -1 \end{cases}$$

Solve it

- Row 1: $a = 2$.
- Row 2: $b = 3 - 2a = -1$.
- Row 3: $b = -1$. It **agrees**.

So $\bar{u} = 2\bar{v} - \bar{w}$, and the set is **dependent**: $\bar{u}$ adds nothing new.

Three equations, two unknowns, and they happened to be **consistent**. That is what “dependent” means.

Example: Change One Number and It Flips

Replace $\bar{u}$ by $\bar{u}' = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$ and ask the same question:

$$a \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} \quad \longleftrightarrow \quad \begin{cases} a = 2 \\ 2a + b = 3 \\ b = 0 \end{cases}$$

Solve it

- Rows 1 and 2 still force $a = 2$, $b = -1$.
- Row 3 demands $b = 0$: a **contradiction**.
- No weights exist. With $\bar{v}$, $\bar{w}$ not multiples of each other, the set is **independent**.

The takeaway

- Testing for dependence **is** solving a system.
- A solution is the redundancy; **no solution** means independent.
- $\bar{u}$ lay in the plane of $\bar{v}$ and $\bar{w}$; $\bar{u}'$ pokes out of it.

Why Dependence Is a Problem

A spare vector sounds harmless. It is not.

- **No unique answer.** If $\bar{u} = 2\bar{v} - \bar{w}$, every vector in the span of $\{\bar{u}, \bar{v}, \bar{w}\}$ is a linear combination of them in **infinitely many** ways.
- **Singular matrices.** Dependent columns make a matrix (next unit) with no inverse; `np.linalg.solve` refuses.
- **Nearly dependent is almost as bad.** Tiny changes in the data swing the weights wildly, and in floating point (Unit 1-5) you get huge coefficients, `inf`, or `NaN`.

Running example: collinear patient features

- Record each patient's height in inches **and** in centimetres: one column is $2.54\times$ the other.
- A regression cannot decide which one deserves the credit, so its coefficients go unstable.
- Fix: drop or combine the redundant column; PCA automates that.

Basis and Change of Basis

Basis and Dimension

Definition

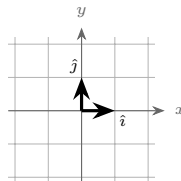
A **basis** is a set of vectors that is **independent** (nothing redundant) and **spanning** (reaches everything).

The **dimension** of the space is the number of vectors in a basis; every basis has the same number.

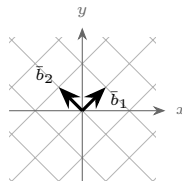
What to know

- **Minimal and complete:** drop one, you miss something; add one, it is redundant.
- **Not unique:** all four at right are bases. The first two are **orthonormal**, the nicest kind.

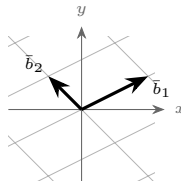
Four bases of $\mathbb{R}^2$, so $\mathbb{R}^2$ has dimension 2.
Faint lines: the integer combinations $a\bar{b}_1 + b\bar{b}_2$, a grid over the **whole** plane.



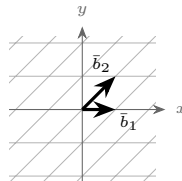
Standard: $\hat{i}, \hat{j}$



Rotated 45°



Skewed: $\begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix}$



Slanted: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Change of Basis: the Same Point, Different Coordinates

Coordinates are the **weights** on the basis vectors: $(2, 4)$ means $2\hat{i} + 4\hat{j}$.

Rotated orthonormal axes

$$\hat{b}_1 = \begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix}, \quad \hat{b}_2 = \begin{bmatrix} -0.6 \\ 0.8 \end{bmatrix}$$

New weights: projections

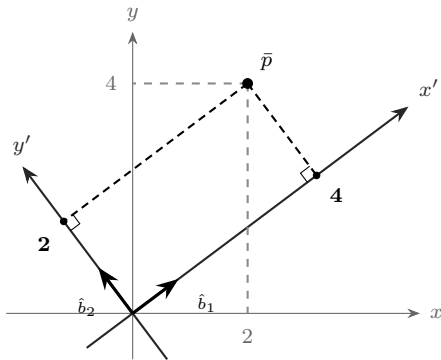
$$c_1 = \bar{p} \cdot \hat{b}_1 = (2)(0.8) + (4)(0.6) = 4$$

$$c_2 = \bar{p} \cdot \hat{b}_2 = (2)(-0.6) + (4)(0.8) = 2$$

Not orthonormal? Solve $c_1\bar{b}_1 + c_2\bar{b}_2 = \bar{p}$ instead.

$$\text{Check: } 4\hat{b}_1 + 2\hat{b}_2 = \begin{bmatrix} 3.2 \\ 2.4 \end{bmatrix} + \begin{bmatrix} -1.2 \\ 1.6 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}. \quad \text{Same point, new address.}$$

Notebook U2-2_Vectors-5_IndependenceAndBasis, §4, the same point read off two grids



Gray dashes: xy address $(2, 4)$
Black dashes: $x'y'$ address $(4, 2)$

Exercise: Span, Independence, and Basis

Try it

- ① Is $\begin{bmatrix} 5 \\ 5 \end{bmatrix}$ in the span of $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$? Is $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$?
- ② Are $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 6 \end{bmatrix}$ independent? What do they span?
- ③ Are $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ independent? Answer before computing.
- ④ Write $\begin{bmatrix} 4 \\ 0 \end{bmatrix}$ in the basis $\bar{b}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\bar{b}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

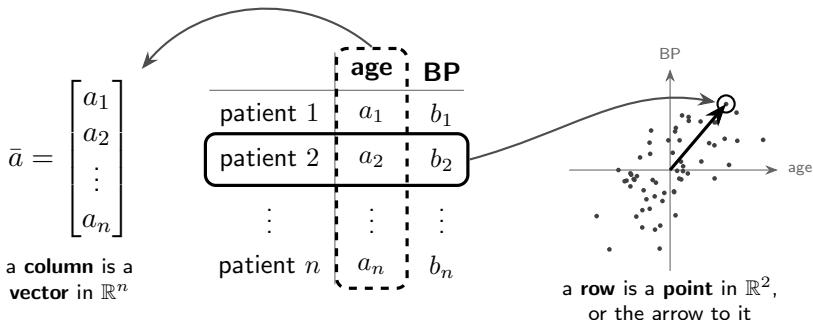
Exercise: Solution

Answers

- ① **Yes** for the first: $\begin{bmatrix} 5 \\ 5 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. **No** for the second: the span of one vector is the line $y = x$, and $(1, 2)$ is not on it. There is no c with $c = 1$ and $c = 2$ at once.
- ② **Dependent**, since the second is 2 times the first. They span only the line through the origin in the direction $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$, not the plane.
- ③ **Dependent**, and no computation is needed: three vectors in $\mathbb{R}^2$ are always dependent, because a basis for $\mathbb{R}^2$ has only two vectors. The first two already span the plane, so the third has to be a combination of them.
- ④ Solve $c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$, that is $c_1 + c_2 = 4$ and $c_1 - c_2 = 0$. Adding gives $2c_1 = 4$, so $c_1 = c_2 = 2$: the coordinates are $(2, 2)$ in the $\bar{b}$ basis. (Note $\bar{b}_1 \cdot \bar{b}_2 = 0$, so this basis is orthogonal, just rotated from the standard one.)

Why Anyone Would Want a Different Basis

Running example: n patients, each with a standardized age a_i and blood pressure b_i . One table, two pictures:

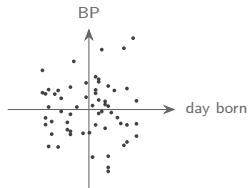


Dependence is about the columns

Is $\bar{b}$ a multiple of $\bar{a}$? No, so the columns are **independent**. But age and BP rise together, so in $\mathbb{R}^n$ the two columns **lean the same way**: the angle between them is well under 90° .

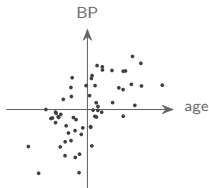
Three Tables, Three Clouds

Three tables like the last one, all standardized (so units drop out):



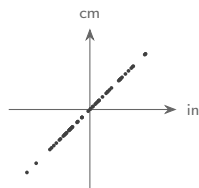
Unrelated

Day of the month born, and BP.
Columns: orthogonal.
Two patients: any angle; cosine similarity from -1 to 1 .



Partly aligned

Age and BP.
Columns: independent, but lean the same way.
Two patients: often near parallel or opposite.



Dependent

Height in inches and in cm.
Columns: dependent ($\text{cm} = 2.54 \times \text{in}$).
Two patients: parallel or opposite; cosine similarity ± 1 .

Rows and columns move together

Cosine similarity of two patients compares **rows**; dependence compares **columns**. The more dependent the columns, the closer every row lies to one line.

How Dependent? Pearson Correlation

Pearson correlation

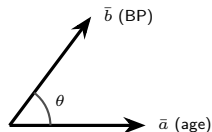
Let a_i, b_i be patient i 's age and BP, **centered** (mean subtracted). Then

$$r = \frac{\sum_{i=1}^n a_i b_i}{\sqrt{\sum_{i=1}^n a_i^2} \sqrt{\sum_{i=1}^n b_i^2}} = \frac{\bar{a} \cdot \bar{b}}{\|\bar{a}\| \|\bar{b}\|} = \cos \theta.$$

A sum over the **points** is a dot product of the **columns**: r is the cosine of the angle θ between $\bar{a}$ and $\bar{b}$ in $\mathbb{R}^n$.

Reading r

- $r = \pm 1$: columns **dependent**; points on a line.
- $r = 0$: columns **orthogonal**; no linear trend.
- The three clouds: about 0, about 0.6, $r = 1$.

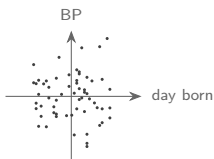


The age and BP columns, in the plane they span: $\cos \theta = r$.

r^2 : Is a Feature Redundant?

Square it

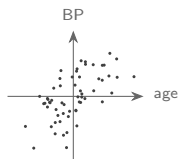
r^2 runs from 0 to 1 and ignores sign ($r = -1$ is as dependent as $r = 1$): the share of one column's spread that a straight line through the other explains.



$$r^2 \approx 0$$

Day born and BP.

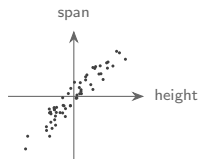
Unrelated: good.



$$r^2 \approx 0.36$$

Age and BP.

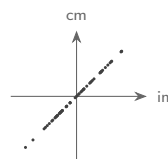
Partly shared.



$$r^2 \approx 0.90$$

Height and arm span.

Nearly dependent:
still bad.



$$r^2 = 1$$

Height, in and cm.

Dependent: bad.

$r^2 = 0$: **good**, all new information

$r^2 = 1$: **bad**, redundant

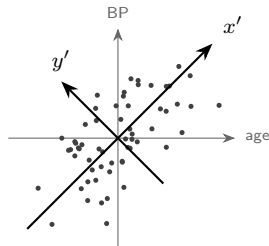
Close to 1 is nearly as bad as 1: regression weights still swing wildly.

Careful: this is r^2 between two **features**. For a model's predictions against the truth, a **high** R^2 is good.

The Fix: Axes Along the Cloud

Change the basis

- Line the axes up with the cloud: x' carries most of the variation.
- y' carries much less, so **drop** it: two correlated columns become **one**, keeping most of the information.



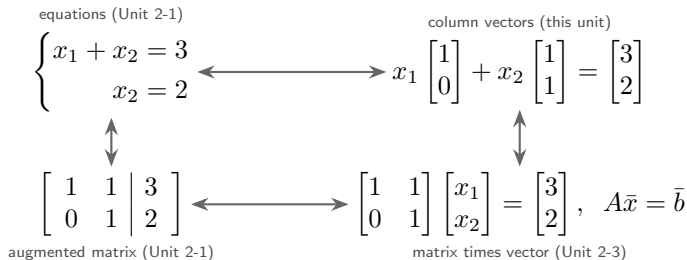
Age and BP, axes along the cloud.

That is PCA

- **Principal component analysis**, this unit's capstone, is a change of basis **chosen well**.
- Its new columns are **uncorrelated**, $r = 0$: the overlap between age and BP is gone.
- Which basis? **Eigenvectors** answer that, in §2.6.
- Same idea as compressing images and audio.

Looking Ahead: Where Matrices Come From

One System, Four Ways to Write It



Two ways to read it

- **Row picture**, read **across**: each equation is a line; the solution is where they cross.
- **Column picture**, read **down**: the unknowns weight the columns; the solution is the **weights** of the linear combination.

Where this is going: the columns side by side are a **matrix** A ; Unit 2-3 defines $A\bar{x}$ as those columns weighted by $\bar{x}$.

What we covered: vectors, operations, angles, and similarity

- ① A **vector** has magnitude and direction. We draw it as an arrow and write it as a list of numbers, a **column** by default or a **row**; the two shapes differ, and Unit 2-3 says how.
- ② A **point** is a location, a vector is a **displacement**; a vector may be slid anywhere without changing.
- ③ A vector with n components lives in $\mathbb{R}^n$, and **one row of a data table is one such vector**.
- ④ **Magnitude** $\|\vec{v}\| = \sqrt{v_1^2 + \cdots + v_n^2}$ is the Pythagorean hypotenuse.
- ⑤ **Addition** is head to tail, **scaling** stays on the vector's line, **subtraction** adds the reverse, and $\hat{v} = \vec{v}/\|\vec{v}\|$ is the **unit vector**.
- ⑥ **Distance** is $\|\vec{v} - \vec{w}\|$; it is scale-sensitive, so standardize first.
- ⑦ The **dot product** $\vec{v} \cdot \vec{w} = \sum v_i w_i$ multiplies matching components and adds; it returns a **number**, not a vector.
- ⑧ $\cos \theta = \frac{\vec{v} \cdot \vec{w}}{\|\vec{v}\| \|\vec{w}\|}$ gives the **angle**; **cosine similarity** ignores size.

Review

What we covered: projection, span, basis, and the look ahead

- ⑨ A dot product of 0 means **orthogonal**, and **projection** is the shadow of one vector on another.
- ⑩ A **linear combination** scales and adds; the **span** is everything reachable that way, and finding the weights is **solving a system** from Unit 2-1.
- ⑪ **Dependent** vectors are redundant (this is **collinearity**); independent ones each add a new direction.
- ⑫ A **basis** is independent and spanning, and its size is the **dimension**; **change of basis** re-describes the same point in different coordinates, which is the idea PCA is built on.
- ⑬ A system is also a **linear combination of column vectors**; collecting those columns into a grid gives a **matrix**, the subject of Unit 2-3.

Notebooks: U2-2_Vectors-1_ArrowsAndOps, -2_DotProductAndAngle, -3_ProjectionAndSimilarity, -4_LinearCombinationsAndSpan, -5_IndependenceAndBasis. **Previous:** Unit 2-1, Systems. **Next:** Unit 2-3, Matrices.